Simultaneous Equations & Inequalities

ANSWER KEY

Linear simultaneous equations

1 Solve the system: $3x + 2y = 16$ and $x - y = 2$.

From the second equation, $x = y + 2$. Substitute: $3(y + 2) + 2y = 16$, so $3y + 6 + 2y = 16$, giving $5y = 10$, $y = 2$. Then $x = 4$.

2 Solve the system: $4x - 3y = 1$ and $2x + y = 9$.

From the second equation, $y = 9 - 2x$. Substitute: $4x - 3(9 - 2x) = 1$, so $4x - 27 + 6x = 1$, giving $10x = 28$, $x = 2.8$. Then $y = 9 - 5.6 = 3.4$.

3 Solve the system: $3x + 4y = 18$ and $x - 2y = -4$.

From the second equation, $x = 2y - 4$. Substitute: $3(2y - 4) + 4y = 18$, so $6y - 12 + 4y = 18$, giving $10y = 30$, $y = 3$. Then $x = 2(3) - 4 = 2$.

4 For the system $x + 3y = 11$ and $2x - y = 1$:

a Solve for x and y. From the second equation, $y = 2x - 1$. Substitute: $x + 3(2x - 1) = 11$, so $7x - 3 = 11$, giving $x = 2$, $y = 3$.

b State whether the point of intersection lies in the first quadrant. Since both $x = 2 > 0$ and $y = 3 > 0$, yes — the point $(2, 3)$ lies in the first quadrant.

Solving a linear and a quadratic equation simultaneously

5 Solve the system: $y = x + 1$ and $y = x^2 - 5$.

Set equal: $x + 1 = x^2 - 5$, so $x^2 - x - 6 = 0$, factoring to $(x - 3)(x + 2) = 0$, giving $x = 3$ or $x = -2$. Corresponding y-values: $(3, 4)$ and $(-2, -1)$.

6 Solve the system: $2x - y = 1$ and $x^2 + y^2 = 29$.

From the first equation, $y = 2x - 1$. Substitute: $x^2 + (2x - 1)^2 = 29$, so $x^2 + 4x^2 - 4x + 1 = 29$, giving $5x^2 - 4x - 28 = 0$. Using the quadratic formula: $x = \frac{4 \pm \sqrt{16 + 560}}{10} = \frac{4 \pm 24}{10}$, so $x = 2.8$ or $x = -2$. Corresponding y-values: $(2.8, 4.6)$ and $(-2, -5)$.

7 Solve the system: $y = 2x^2$ and $y = x + 1$.

Set equal: $2x^2 = x + 1$, so $2x^2 - x - 1 = 0$, factoring to $(2x + 1)(x - 1) = 0$, giving $x = -\frac{1}{2}$ or $x = 1$. Corresponding y-values: $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and $(1, 2)$.

Using the discriminant to determine intersections

8 Find the values of k for which the line $y = kx$ is tangent to the curve $y = x^2 + 4$ (i.e. meets it at exactly one point).

Set equal: $kx = x^2 + 4$, so $x^2 - kx + 4 = 0$. For tangency, the discriminant is zero: $k^2 - 16 = 0$, giving $k = 4$ or $k = -4$.

9 Find the values of c for which the line $y = 3x + c$ does not intersect the curve $y = x^2 + 2x + 5$.

Set equal: $3x + c = x^2 + 2x + 5$, so $x^2 - x + (5 - c) = 0$. For no intersection, the discriminant must be negative: $1 - 4(5 - c) < 0$, so $1 - 20 + 4c < 0$, giving $4c < 19$, so $c < \frac{19}{4}$.

10 Find the values of m for which the line $y = mx + 2$ intersects the curve $y = x^2 + 3x + 7$ at two distinct points.

Set equal: $mx + 2 = x^2 + 3x + 7$, so $x^2 + (3 - m)x + 5 = 0$. For two distinct real intersections, the discriminant must be positive: $(3 - m)^2 - 20 > 0$, so $(3 - m)^2 > 20$, giving $3 - m > 2\sqrt{5}$ or $3 - m < -2\sqrt{5}$. So $m < 3 - 2\sqrt{5}$ or $m > 3 + 2\sqrt{5}$.

Solving linear inequalities

11 Solve $5x - 3 > 2x + 9$.

$3x > 12$, so $x > 4$.

12 Solve $-3x + 7 \leq 1$, being careful with the direction of the inequality.

$-3x \leq -6$. Dividing by -3 flips the inequality: $x \geq 2$.

13 Solve $2(x - 3) + 5 \geq 3x - 4$.

$2x - 6 + 5 \geq 3x - 4$, so $2x - 1 \geq 3x - 4$, giving $-x \geq -3$, so $x \leq 3$.

14 Solve $4 - 2x < 10$.

$4 - 2x < 10$: $-2x < 6$. Dividing by -2 flips the inequality: $x > -3$.

Solving quadratic inequalities

15 Solve $x^2 - 5x + 6 < 0$, and represent the solution on a number line.

Factor: $(x - 2)(x - 3) < 0$. This is negative between the roots (since the parabola opens upward and dips below zero between its two zeros): $2 < x < 3$.

16 Solve $x^2 \geq 9$.

$x^2 - 9 \geq 0$, so $(x - 3)(x + 3) \geq 0$. This holds when $x \leq -3$ or $x \geq 3$.

17 Solve $2x^2 - 3x - 2 \leq 0$.

Factor: $2x^2 - 3x - 2 = (2x + 1)(x - 2)$. This is ≤ 0 between the roots (since the parabola opens upward): $-\frac{1}{2} \leq x \leq 2$.

18 For the inequality $x^2 - 4x - 5 \geq 0$:

a Factor the quadratic and find its roots. $(x-5)(x+1)=0$, so the roots are $x=5$ and $x=-1$.

b State the solution set for the inequality. Since the parabola opens upward, the inequality holds outside the roots: $x \leq -1$ or $x \geq 5$.

Combining a quadratic inequality with a linear constraint

19 Solve the compound inequality $x^2 - 2x - 8 \leq 0$ and $x > -1$ simultaneously, giving the combined solution set.

Factor the quadratic: $(x-4)(x+2) \leq 0$, giving $-2 \leq x \leq 4$. Combining with $x > -1$: the intersection is $-1 < x \leq 4$.

20 Solve the compound inequality $x^2 - 7x + 10 \leq 0$ and $x < 4$ simultaneously, giving the combined solution set.

Factor the quadratic: $(x-2)(x-5) \leq 0$, giving $2 \leq x \leq 5$. Combining with $x < 4$: the intersection is $2 \leq x < 4$.

21 Solve the compound inequality $x^2 - 9 \leq 0$ and $x \geq -2$ simultaneously, giving the combined solution set.

Factor the quadratic: $(x-3)(x+3) \leq 0$, giving $-3 \leq x \leq 3$. Combining with $x \geq -2$: the intersection is $-2 \leq x \leq 3$.

Simultaneous equations from a real-world constraint

22 A rectangle has perimeter 34 and diagonal length 13. Find its length and width.

Let length $=l$, width $=w$: $2l + 2w = 34$, so $l + w = 17$. Diagonal: $l^2 + w^2 = 169$. From the first equation, $w = 17 - l$. Substitute: $l^2 + (17-l)^2 = 169$, so $l^2 + 289 - 34l + l^2 = 169$, giving $2l^2 - 34l + 120 = 0$, or $l^2 - 17l + 60 = 0$. Factor: $(l-5)(l-12) = 0$, so $l = 12, w = 5$ or $l = 5, w = 12$ (the same rectangle either way).

23 Two numbers have a sum of 20 and a product of 91. Find the two numbers.

Let the numbers be x and y: $x + y = 20$ and $xy = 91$. From the first equation, $y = 20 - x$. Substitute: $x(20-x) = 91$, so $x^2 - 20x + 91 = 0$. Using the quadratic formula: $x = \frac{20 \pm \sqrt{400-364}}{2} = \frac{20 \pm 6}{2}$, giving $x = 13$ or $x = 7$. The two numbers are 7 and 13.

24 Two positive numbers have a difference of 5 and a product of 84. Find the two numbers.

Let the numbers be x (larger) and y (smaller): $x - y = 5$ and $xy = 84$. From the first equation, $x = y + 5$. Substitute: $(y+5)y = 84$, so $y^2 + 5y - 84 = 0$. Using the quadratic formula: $y = \frac{-5 \pm \sqrt{25+336}}{2} = \frac{-5 \pm 19}{2}$, giving $y = 7$ (taking the positive root). Then $x = 12$. The two numbers are 7 and 12.

A well-designed word problem: break-even analysis

25 This question has two parts. A company's costs are modeled by $C(x) = 500 + 8x$ dollars and revenue by $R(x) = 20x - 0.02x^2$ dollars, where x is the number of units sold. (a) Set up and solve the inequality $R(x) > C(x)$ to find the range of production levels for which the company makes a profit. (b) Explain, in terms of the model, why profit is not simply increasing forever as x grows.

(a) $20x - 0.02x^2 > 500 + 8x$, so $-0.02x^2 + 12x - 500 > 0$, or (multiplying by -50 and flipping the inequality) $x^2 - 600x + 25000 < 0$. Using the quadratic formula on the boundary equation $x^2 - 600x + 25000 = 0$: $x = \frac{600 \pm \sqrt{360000 - 100000}}{2} = \frac{600 \pm \sqrt{260000}}{2} \approx \frac{600 \pm 509.9}{2}$, giving $x \approx 45.05$ or $x \approx 554.95$. The company profits for roughly $45 < x < 555$ units. (b) The revenue function $R(x) = 20x - 0.02x^2$ is itself a downward-opening parabola (due to the $-0.02x^2$ term), meaning revenue eventually decreases as production grows too large (e.g. due to market saturation or price drops needed to sell more units) — so profit, being revenue minus a steadily increasing cost, eventually falls back below zero at very high production levels too.